

QUBIT TOUCHDOWN · Facilitator Guide

A quantum football board game · Grades 6–8

Players	2 players per board (or 2 teams on a big floor board)
Time	40–60 minutes
Big idea	Quantum gates move a qubit between states, like plays move a football.

Your job as the facilitator

You are the game guide. You do not need to be a quantum expert. Your job is to keep the game moving, read the rules out loud, and ask good questions. It is fine to say "Let's find out together."

What the game is about

A football token is a **qubit**, the basic piece of a quantum computer. The board has 6 spots: **0, 1, +, −, i and −i**. A normal computer bit can only be 0 or 1. A qubit can be at other spots too. Action cards are **quantum gates**. Each gate moves the football along the lines on the board. Get the football into your opponent's end zone (the + or − spot on their side) to score a touchdown.

What you need

- The Qubit Touchdown board (one per pair)
- The 52 action cards (X, Y, Z, I, H, S, √X and Measurement)
- A football token
- A "binary die": any die where you call low numbers 0 and high numbers 1, or a coin (heads = 0, tails = 1)
- Optional: the node tokens, to make a giant floor board with tape

Get ready

1. Put the board between the two players. Each player has an end zone facing them.
2. Shuffle the 52 cards. Give each player 4 cards. Put the other 44 face-down as a draw pile. Choose a spot for the discard pile.
3. One player **kicks off**: roll the binary die and put the football on 0 or 1 to match.

Run the game, step by step

> **Say:** “The football is a qubit. Your cards are quantum gates. Use them to move the qubit into the end zone.”

1. The other player plays one card from their hand.
2. **Gate card:** move the football along the **white lines** for that gate. Ignore the green lines. If there is no line for that gate from where the football is, it stays put.
3. **Measurement card:** if the football is at 0 or 1, nothing happens. If it is anywhere else, roll the binary die and move the football to 0 or 1.
4. Draw a new card so you have 4 again.
5. If the football is in an end zone (+ or −), the player whose opponent's end zone it is scores a touchdown. Then kick off again.

6 Take turns until all 52 cards are played. Most touchdowns wins.

Example moves from 0
H moves the football to +. √X moves it to −i. X or Y moves it to 1. Z, I and S keep it at 0.
Lines go both ways, but arrows go one way only. √X goes from −i to 1, not back to 0.

Grades 6–8 extra rule (suggested)

Before you play a card, say your prediction out loud: "I am at 0, and I think H will move me to +." A correct prediction earns 1 bonus point. Final score = touchdowns + prediction points. (This mixes the "Quantum Quarterback" rule from the grades 3–5 version with the full card set.)

Watch for these mistakes

- Following the green lines. Only the white lines count for moves.
- Moving backward along a one-way arrow.
- Using Measurement at 0 or 1 and expecting a change. Measuring 0 always gives 0.

Questions to ask

- Why does measuring at + give 0 or 1 with a 50/50 chance, but measuring at 0 always gives 0?
- Which card is most useful near your opponent's end zone? Which is risky?
- Can you find two cards that undo each other?
- How is a qubit different from a normal 0-or-1 bit?

The quantum words

In the game	In a quantum computer
Football token	A qubit
Spots 0 and 1	The two states a normal bit can have
Spots +, −, i and −i	Superpositions: half 0 and half 1 when measured
Gate cards	Quantum gates (rotations of the qubit)
Measurement card and kick-off	Measuring the qubit
Back of the card (a ball)	The Bloch sphere: a map of every qubit state

Wrap up (say this)

> **Say:** “Every move you made was a real quantum gate. A quantum program is a list of gates, just like the list of cards you played.”

Game by Kristina Armbruster, Mary Duda and Thomas Wong. Teacher variations from the QCaMP slide deck in this folder.